



# A spectral technique for coloring random 3-colorable graphs (Preliminary Version)

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## Abstract

Let  $G(3n, p, 3)$  be a random 3-colorable graph on a set of  $3n$  vertices generated as follows. First, split the vertices arbitrarily into three equal color classes and then choose every pair of vertices of distinct color classes, randomly and independently, to be an edge with probability  $p$ . We describe a polynomial time algorithm that finds a proper 3-coloring of  $G(3n, p, 3)$  with high probability, whenever  $p \geq c/n$ , where  $c$  is a sufficiently large absolute constant. This settles a problem of Blum and Spencer, who asked if one can design an algorithm that works almost surely for  $p \geq \text{polylog}(n)/n$ . The algorithm can be extended to produce optimal  $k$ -colorings of random  $k$ -colorable graphs in a similar model, as well as in various related models.

## 1 Introduction

A vertex coloring of a graph  $G$  is *proper* if no adjacent vertices receive the same color. The *chromatic*

number  $\chi(G)$  of  $G$  is the minimum number of colors in a proper vertex coloring of it. The problem of determining or estimating this parameter received a considerable amount of attention in Combinatorics and in Theoretical Computer Science, as several scheduling problems are naturally formulated as graph coloring problems. It is well known that the problem of coloring properly a graph of chromatic number  $k$  with  $k$  colors is *NP*-hard, even for any fixed  $k \geq 3$ , and it is therefore unlikely that there are efficient algorithms for coloring optimally an arbitrary 3-chromatic input graph.

On the other hand, various researchers noticed that *random*  $k$ -colorable graphs are usually easy to color optimally. Polynomial time algorithms that color optimally random  $k$ -colorable graphs, for every fixed  $k$ , with high probability have been developed by Kucera [12], by Turner [15] and by Dyer and Frieze [7], where the last paper provides an algorithm whose average running time over all  $k$ -colorable graphs on  $n$  vertices is polynomial. Note, however, that most  $k$ -colorable graphs are quite dense, and hence easy to color. In fact, in a typical  $k$ -colorable graph, the number of common neighbors of any pair of vertices with the same color exceeds considerably that of any pair of vertices of distinct colors, and hence a simple coloring algorithm based on this fact already works with high probability. It is more difficult to color sparser random  $k$ -colorable graphs. A precise model for generating sparse random  $k$ -colorable graphs is described in the next subsection, where the sparsity is governed by a parameter  $p$  that specifies the edge probability. Petford and Welsh [13] suggested a randomized heuristic for 3-coloring random 3-colorable graphs and supplied experimental evidence that it works for most edge probabilities. Blum and Spencer [4]

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(see also [2] for some related results) designed a polynomial algorithm and proved that it colors optimally with high probability random 3-colorable graphs on  $n$  vertices with edge probability  $p$  provided  $p \geq n^\epsilon/n$ , for some arbitrarily small but fixed  $\epsilon > 0$ . Their algorithm is based on a path counting technique, and can be viewed as a natural generalization of the simple algorithm based on counting common neighbors (that counts paths of length 2), mentioned above.

Our main result here is a polynomial time algorithm that works for sparser random 3-colorable graphs. If the edge probability  $p$  satisfies  $p \geq c/n$ , where  $c$  is a sufficiently large absolute constant, the algorithm colors optimally the corresponding random 3-colorable graph with high probability. This settles a problem of Blum and Spencer [4], who asked if one can design an algorithm that works almost surely for  $p \geq \text{polylog}(n)/n$ . (Here, and in what follows, *almost surely* always means: with probability that approaches 1 as  $n$  tends to infinity). The algorithm uses the spectral properties of the graph and is based on the fact that almost surely a rather accurate approximation of the color classes can be read from the eigenvectors corresponding to the smallest two eigenvalues of the adjacency matrix of a large subgraph. This approximation can then be improved to yield a proper coloring.

The algorithm can be easily extended to the case of  $k$ -colorable graphs, for any fixed  $k$ , and to various models of random *regular* 3-colorable graphs of *constant* degree.

## 1.1 The model

There are several possible models for random  $k$ -colorable graphs. See [7] for some of these models and the relation between them. Our results hold for most of these models, but it is convenient to focus on one, which will simplify the presentation. Let  $V$  be a fixed set of  $kn$  labelled vertices. For a real  $p = p(n)$ , let  $G_{kn,p,k}$  be the random graph on the set of vertices  $V$  obtained as follows; first, split the vertices of  $V$  arbitrarily into  $k$  color classes  $V_1, \dots, V_k$ , each of cardinality  $n$ . Next, for each  $u$  and  $v$  that lie in distinct color classes, choose  $uv$  to be an edge, randomly and independently, with probability  $p$ . The input to our algorithm is a graph  $G_{kn,p,k}$  obtained as above, and the algorithm suc-

ceeds to color it if it finds a proper  $k$  coloring. Here we are interested in fixed  $k \geq 3$  and large  $n$ . We say that an algorithm colors  $G_{kn,p,k}$  *almost surely* if the probability that a randomly chosen graph as above is properly colored by the algorithm tends to one as  $n$  tends to infinity. Note that we consider here deterministic algorithms, and the above statement means that the algorithm succeeds to color almost all random graphs generated as above.

A closely related model to the one given above is the model in which we do not insist that the color classes have equal sizes. In this model one first splits the set of vertices into  $k$  disjoint color classes by letting each vertex choose its color randomly, independently and uniformly among the  $k$  possibilities. Next, one chooses every pair of vertices of distinct color classes to be an edge with probability  $p$ . All our results hold for both models, and we focus on the first one as it is more convenient. To simplify the presentation, we restrict our attention to the case  $k = 3$  of three colorable graphs, since the results for this case easily extend to every *fixed*  $k$ . In addition, we make no attempt to optimize the constants and assume, whenever this is needed, that  $c$  is a sufficiently large constant, and the number of vertices  $3n$  is sufficiently large.

## 1.2 The algorithm

Here is a description of the algorithm, which consists of three phases. Given a graph  $G = G_{3n,p,3} = (V, E)$ , define  $d = pn$ . Let  $G' = (V, E')$  be the graph obtained from  $G$  by deleting all edges adjacent to a vertex of degree greater than  $5d$ . Denote by  $A$  the adjacency matrix of  $G'$ , i.e., the  $3n$  by  $3n$  matrix  $(a_{uv})_{u,v \in V}$  defined by  $a_{uv} = 1$  if  $uv \in E'$  and  $a_{uv} = 0$  otherwise. It is well known that since  $A$  is symmetric it has real eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{3n}$  and an orthonormal basis of eigenvectors  $e_1, e_2, \dots, e_{3n}$ , where  $Ae_i = \lambda_i e_i$ . The crucial point is that almost surely one can deduce a good approximation of the coloring of  $G$  from  $e_{3n-1}$  and  $e_{3n}$ . Note that there are several efficient algorithms to compute the eigenvalues and the eigenvectors of symmetric matrices (cf., e.g., [14]) and hence  $e_{3n-1}$  and  $e_{3n}$  can certainly be calculated in polynomial time. For the rest of the algorithm, we will deal with  $G$  rather than  $G'$ .

Let  $t$  be a non-zero linear combination of  $e_{3n-1}$  and  $e_{3n}$  whose median is zero, that is, the number

of positive components of  $t$  as well as the number of its negative components are both at most  $3n/2$ . (It is easy to see that such a combination always exists and can be found efficiently.) Suppose also that  $t$  is normalized so that its  $l_2$ -norm is  $\sqrt{2n}$ . Define  $V_1^0 = \{u \in V : t_u > 1/2\}$ ,  $V_2^0 = \{u \in V : t_u < -1/2\}$ , and  $V_3^0 = \{u \in V : |t_u| \leq 1/2\}$ . This is an approximation for the coloring, which will be improved in the second phase by iterations, and then in the third phase to obtain a proper 3-coloring.

In iteration  $i$  of the second phase,  $0 < i \leq q = \lceil \log n \rceil$ , construct the color classes  $V_1^i, V_2^i$  and  $V_3^i$  as follows. For every vertex  $v$  of  $G$ , let  $N(v)$  denote the set of all its neighbors in  $G$ . In the  $i$ -th iteration, color  $v$  by the least popular color of its neighbors in the previous iteration. That is, put  $v$  in  $V_j^i$  if  $|N(v) \cap V_j^{i-1}|$  is the minimum among the three quantities  $|N(v) \cap V_l^{i-1}|$ , ( $l = 1, 2, 3$ ), where equalities are broken arbitrarily. We will show that the three sets  $V_j^q$  correctly color all but  $n2^{-\Omega(d)}$  vertices.

The third phase consists of two stages. First, repeatedly uncolor every vertex colored  $j$  that has less than  $d/2$  neighbors (in  $G$ ) colored  $l$ , for some  $l \in \{1, 2, 3\} - \{j\}$ . Then, if the graph induced on the set of uncolored vertices has a connected component of size larger than  $\log_3 n$ , the algorithm fails. Otherwise, find a coloring of every component consistent with the rest of the graph using brute force exhaustive search. If the algorithm cannot find such a coloring, it fails.

Our main result is the following.

**Theorem 1.1** *If  $p > c/n$ , where  $c$  is a sufficiently large constant, the algorithm produces a proper 3-coloring of  $G$  with probability  $1 - o(1)$ .*

The proof of this theorem is presented in the next two sections. We first show, in Section 2, that almost surely the largest eigenvalue of  $G'$  is at least  $(1 - 2^{-\Omega(d)})2d$ , its two smallest eigenvalues are at most  $-(1 - 2^{-\Omega(d)})d$  and all other eigenvalues are in absolute value  $O(\sqrt{d})$ . This is done by a proper modification of techniques developed by Friedman, Kahn and Szemerédi in [10]. Next we show in Section 3 that this implies that each of the two eigenvectors corresponding to the two smallest eigenvalues is close to a vector which is a constant on every color class, where the sum of these three constants is zero. This suffices to show that the sets  $V_j^0$  form

a reasonably good approximation to the coloring of  $G$ , with high probability.

Theorem 1.1 can then be proved by applying the expansion properties of the graph  $G$  (that hold almost surely) to show that the iteration process above converges quickly to a proper coloring of a large subgraph  $H$  of  $G$ . The uncoloring procedure will uncolor all vertices which are wrongly colored, but will not affect the subgraph  $H$ . We then conclude by showing that the largest connected component of the induced subgraph of  $G$  on  $V - H$  is of logarithmic size almost surely, thereby showing that the brute-force search on the set of uncolored vertices terminates in polynomial time. Section 4 contains some concluding remarks together with possible extensions and results for related models of random graphs.

## 2 Bounding the eigenvalues

Let  $G = G_{3n,p,3} = (V, E)$  be a random 3-colorable graph generated according to the model described above. Let  $G'$  be the graph obtained from  $G$  by deleting all edges adjacent to vertices of degree greater than  $5d$ , and let  $A$  be the adjacency matrix of  $G'$ . Denote by  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{3n}$  the eigenvalues of  $A$ , and by  $e_1, e_2, \dots, e_{3n}$  the corresponding eigenvectors, chosen so that they form an orthonormal basis of  $R^{3n}$ .

In this section we prove the following statement.

**Proposition 2.1** *In the above notation, almost surely,*

- (i)  $\lambda_1 \geq (1 - 2^{-\Omega(d)})2d$ ,
- (ii)  $\lambda_{3n} \leq \lambda_{3n-1} \leq -(1 - 2^{-\Omega(d)})d$  and
- (iii)  $|\lambda_i| \leq O(\sqrt{d})$  for all  $2 \leq i \leq 3n - 2$ .

**Remark.** One can show that, when  $p = o(\log n/n)$ , it is necessary to deal with the eigenvectors of  $G'$  rather than those of  $G$ , since the graph  $G$  is likely to contain many vertices of degree  $\gg d$ , and in this case the assertion of (iii) does not hold for the eigenvalues of  $G$ .

We need the following lemma, whose simple derivation from the Chernoff bounds is omitted.

**Lemma 2.2** *There exists a constant  $\beta > 0$  such that, almost surely, for any subset  $X$  of  $2^{-\beta d}n$  vertices,  $e(X, V) \leq 5d|X|$ , where  $e(X, V)$  is the number of edges  $(u, v)$ , with  $u \in X$ .*

**Proof of Proposition 2.1 (Outline).**

Parts (i) and (ii) are simple. By the variational definition of eigenvalues,  $\lambda_1$  is simply the maximum of  $x^t Ax / (x^t x)$  where the maximum is taken over all nonzero vectors  $x$ . Therefore, by taking  $x$  to be the all 1 vector we obtain the well known result that  $\lambda_1$  is at least the average degree of  $G'$ . By the known estimates for Binomial distributions, the average degree of  $G$  is  $(1 + o(1))2d$ . On the other hand, lemma 2.2 can be used to show that  $|E - E'| \leq 2^{-\Omega(d)}n$ , as it easily implies that the number of vertices of degree greater than  $5d$  in each color class of  $G$  is almost surely less than  $2^{-\beta d}n$ . Hence, the average degree of  $G'$  is at least  $(1 - 2^{-\Omega(d)})2d$ . This proves (i).

The proof of (ii) is similar. It is known that

$$\lambda_{3n-1} = \min_F \max_{x \in F, x \neq 0} \frac{x^t Ax}{x^t x},$$

where the minimum is taken over all two dimensional subspaces  $F$  of  $R^{3n}$ . Let  $W_1, W_2$  and  $W_3$  denote the three color classes of vertices of  $G$  and let  $F$  denote the 2-dimensional subspace of all vectors  $x = (x_v : v \in V)$  satisfying  $x_v = \alpha_i$  for all  $v \in W_i$ , where  $\alpha_1 + \alpha_2 + \alpha_3 = 0$ . For  $x$  as above

$$\begin{aligned} x^t Ax &= 2\alpha_1\alpha_2e(W_1, W_2) + 2\alpha_2\alpha_3e(W_2, W_3) \\ &\quad + 2\alpha_1\alpha_3e(W_1, W_3), \end{aligned}$$

where  $e(W_i, W_j)$  denotes the number of edges of  $G$  between  $W_i$  and  $W_j$ . Almost surely  $e(W_i, W_j) \geq (1 - 2^{-\Omega(d)})nd$  for all  $1 \leq i < j \leq 3$ , and since  $x^t x = n(\alpha_1^2 + \alpha_2^2 + \alpha_3^2) = -2n(\alpha_1\alpha_2 + \alpha_2\alpha_3 + \alpha_1\alpha_3)$  it follows that  $x^t Ax / (x^t x) \leq -(1 - 2^{-\Omega(d)})d$  almost surely for all  $x \in F$ , implying that  $\lambda_{3n} \leq \lambda_{3n-1} \leq -(1 - 2^{-\Omega(d)})d$ , and establishing (ii).

The proof of (iii) is more complicated. Its assertion for somewhat bigger  $p$  (for example, for  $p \geq \log^6 n/n$ ) can be deduced from the arguments of [9]. To prove it for the graph  $G'$  and  $p \geq c/n$  we use the basic approach of Kahn and Szemerédi in [10], where the authors show that the second largest eigenvalue in absolute value of a random  $d$ -regular graph is almost surely  $O(\sqrt{d})$ . (See also [8] for a different proof.) Since in our case the graph is not regular a few modifications are needed. We sketch the argument below. The full details will appear in the full version of the paper. Our starting point is again the variational definition of the eigenvalues, from which it is easy to deduce that it suffices to show that almost surely the following holds.

**Lemma 2.3** *Let  $S$  be the set of all unit vectors  $x = (x_v : v \in V)$  for which  $\sum_{v \in W_j} x_v = 0$  for  $j = 1, 2, 3$ , then  $|x^t Ax| \leq O(\sqrt{d})$  for all  $x \in S$ .*

The matrix  $A$  consists of nine blocks arising from the partition of its rows and columns according to the classes  $W_j$ . It is clearly sufficient to show that the contribution of each block to the sum  $x^t Ax$  is bounded, in absolute value, by  $O(\sqrt{d})$ . This, together with a simple argument based on  $\epsilon$ -nets (see [10], Proposition 2.1) can be used to show that Lemma 2.3 follows from the following statement.

Fix  $\epsilon > 0$ , say  $\epsilon = 1/2$ , and let  $T$  denote the set of all vectors  $x$  of length  $n$  every coordinate of which is an integral multiple of  $\epsilon/\sqrt{n}$ , where the sum of coordinates is zero and the  $l_2$ -norm is at most 1. Let  $B$  be a random  $n$  by  $n$  matrix with 0, 1 entries, where each entry of  $B$ , randomly and independently, is 1 with probability  $d/n$ .

**Lemma 2.4** *If  $d$  exceeds a sufficiently large absolute constant then almost surely,  $|x^t By| \leq O(\sqrt{d})$  for every  $x, y \in T$  for which  $x_u = 0$  if the corresponding row of  $B$  has more than  $5d$  nonzero entries and  $y_v = 0$  if the corresponding column of  $B$  has more than  $5d$  nonzero entries.*

The last lemma is proved, as in [10], by separately bounding the contribution of terms  $x_u y_v$  with small absolute values and the contribution of similar terms with large absolute values. Specifically, let  $C$  denote the set of all pairs  $(u, v)$  with  $|x_u y_v| \leq \sqrt{d}/n$  and let  $X = \sum_{(u,v) \in C} x_u B(u, v) y_v$ . As in [10] one can show that the absolute value of the expectation of  $X$  is at most  $\sqrt{d}$ . Next one has to show that with high probability  $X$  does not deviate from its expectation by more than  $c\sqrt{d}$ . This is different (and in fact, somewhat easier) than the corresponding result in [10], since here we are dealing with independent random choices. It is convenient to use the following variant of the Chernoff bound.

**Lemma 2.5** *Let  $a_1, \dots, a_m$  be (not necessarily positive) reals, and let  $Z$  be the random variable  $Z = \sum_{i=1}^m \epsilon_i a_i$ , where each  $\epsilon_i$  is chosen, randomly and independently, to be 1 with probability  $p$  and 0 with probability  $1 - p$ . Suppose  $\sum_{i=1}^m a_i^2 \leq D$  and suppose  $|Sa_i| \leq ce^p D$  for some positive constants  $c, S$ . Then  $\text{Prob}[|Z - E(Z)| > S] \leq 2e^{-S^2/(2pe^c D)}$ .*

For the proof, one first proves the following.

**Lemma 2.6** *Let  $c$  be a positive real. Then for every  $x \leq c$ ,*

$$e^x \leq 1 + x + \frac{e^c}{2} x^2.$$

**Proof.** Define  $f(x) = 1 + x + \frac{e^c}{2} x^2 - e^x$ . Then  $f(0) = 0$ ,  $f'(x) = 1 + e^c x - e^x$  and  $f''(x) = e^c - e^x$ . Therefore,  $f''(x) \geq 0$  for all  $x \leq c$  and as  $f'(0) = 0$  this shows that  $f'(x) \leq 0$  for  $x < 0$  and  $f'(x) \geq 0$  for  $c \geq x > 0$ , implying that  $f(x)$  is nonincreasing for  $x \leq 0$  and nondecreasing for  $c \geq x \geq 0$ . Thus  $f(x) \geq 0$  for all  $x \leq c$ , as needed.  $\square$

**Proof of Lemma 2.5.** Define  $\lambda = \frac{S}{e^c p D}$ , then, by assumption,  $\lambda a_i \leq c$  for all  $i$ . Therefore, by the above lemma,

$$\begin{aligned} E(e^{\lambda Z}) &= \prod_{i=1}^m [p e^{\lambda a_i} + (1-p)] = \prod_{i=1}^m [1 + p(e^{\lambda a_i} - 1)] \\ &\leq \prod_{i=1}^m [1 + p(\lambda a_i + \frac{e^c}{2} \lambda^2 a_i^2)] \\ &\leq e^{p\lambda \sum_{i=1}^m a_i + p \frac{e^c}{2} \lambda^2 \sum_{i=1}^m a_i^2} \\ &\leq e^{\lambda E(Z) + \frac{p e^c}{2} \lambda^2 D}. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Prob}[(Z - E(Z)) > S] &= \text{Prob}[e^{\lambda(Z - E(Z))} > e^{\lambda S}] \\ &\leq e^{-\lambda S} E(e^{\lambda(Z - E(Z))}) \leq e^{-\lambda S + \frac{p e^c}{2} \lambda^2 D} = e^{-\frac{S^2}{2 p e^c D}}. \end{aligned}$$

Applying the same argument to the random variable defined with respect to the reals  $-a_i$ , the assertion of the lemma follows.  $\square$

Using Lemma 2.5 it is not difficult to deduce that almost surely the contribution of the pairs in  $C$  to  $|x^t B y|$  is  $O(\sqrt{d})$ . This is because we can simply apply the lemma with  $m = |C|$ , with the  $a_i$ 's being all the terms  $x_u y_v$  where  $(u, v) \in C$ , with  $p = d/n$ ,  $D = 1$  and  $S = c e^c \sqrt{d}$  for some  $c > 0$ . Since here  $|S a_i| \leq c e^c \sqrt{d} \sqrt{d}/n = c e^c p D$ , we conclude that for every fixed vectors  $x$  and  $y$  in  $T$ , the probability that  $X$  deviates from its expectation (which is  $O(\sqrt{d})$ ) by more than  $c e^c \sqrt{d}$  is smaller than  $2e^{-c^2 e^c n/2}$ , and since the cardinality of  $T$  is only  $b^n$  for some absolute constant  $b = b(c)$ , one can choose  $c$  so that  $X$  would never deviate from its expectation by more than  $c e^c \sqrt{d}$ .

The contribution of the terms  $x_u y_v$  whose absolute values exceed  $\sqrt{d}/n$  can be bounded by following the arguments of [10], with a minor modification arising from the fact that the maximum number of ones in a row (or column) of  $B$  can exceed

$d$  (but can never exceed  $5d$  in a row or a column in which the corresponding coordinates  $x_u$  or  $y_v$  are nonzero). This implies the assertion of Lemma 2.4, which implies Lemma 2.3 and part (iii) of the proposition. We omit the details.  $\square$

### 3 The Proof of the main result

In this section we first show that the approximate coloring produced by the algorithm using the eigenvectors  $e_{3n-1}$  and  $e_{3n}$  is rather accurate almost surely. Then we exhibit a large subgraph  $H$  and show that, almost surely, the iterative procedure for improving the coloring colors  $H$  correctly. We then show that the third phase finds a proper coloring of  $G$  in polynomial time, almost surely.

#### 3.1 The properties of the last two eigenvectors

Let  $G = (V, E)$ ,  $A, p, d, \lambda_i, e_i, W_1, W_2, W_3$  be as in Section 2. Let  $x = (x_v : v \in V)$  be the vector defined by  $x_v = 2$  for  $v \in W_1$ , and  $x_v = -1$  otherwise. Let  $y = (y_v : v \in V)$  be the vector defined by  $y_v = 0$  if  $v \in W_1$ ,  $y_v = 1$  if  $v \in W_2$  and  $y_v = -1$  if  $v \in W_3$ . Note that  $x^t y = 0$ , and that both  $\|x\|_2^2$  and  $\|y\|_2^2$  are  $\Theta(n)$ .

**Lemma 3.1** *Almost surely there are two vectors  $\epsilon = (\epsilon_v : v \in V)$  and  $\delta = (\delta_v : v \in V)$ , satisfying  $\|\epsilon\|_2^2 = O(n/d)$  and  $\|\delta\|_2^2 = O(n/d)$  so that  $x - \epsilon$  and  $y - \delta$  are both linear combinations of  $e_{3n-1}$  and  $e_{3n}$ .*

**Proof (outline).** We sketch the proof of existence of  $\delta$  as above. The proof of the existence of  $\epsilon$  is analogous. Let  $y = \sum_{i=1}^{3n} c_i e_i$ . Then  $(A + dI)y = \sum_{i=1}^{3n} c_i (\lambda_i + d) e_i$ , and so

$$\begin{aligned} \|(A + dI)y\|_2^2 &= \sum_{i=1}^{3n} c_i^2 (\lambda_i + d)^2 \quad (1) \\ &\geq \Omega(d^2) \sum_{i=1}^{3n-2} c_i^2, \end{aligned}$$

where the last inequality follows from parts (i) and (iii) of Proposition 2.1. We need the following;

**Claim:** Almost surely:  $\|(A + dI)y\|_2^2 = O(nd)$ .

To prove this claim, observe that it suffices to show that the sum of squares of the coordinates of  $(A + dI)y$  on  $W_1$  is  $O(nd)$  almost surely, as the sums on  $W_2$  and  $W_3$  can be bounded similarly. The expectation of the square of each coordinate of  $(A + dI)y$  is  $O(d)$ , by a standard calculation. Similarly, the expectation of the fourth power of each coordinate of  $(A + dI)y$  is  $O(d^2)$ . Hence, the variance of the square of each coordinate is  $O(d^2)$ . However, the coordinates of  $(A + dI)y$  on  $W_1$  are independent random variables, and hence the variance of the sum of the squares of the  $W_1$  coordinates is equal to the sum of the variances, which is  $O(nd^2)$ . The claim can now be deduced from Chebyshev's Inequality. We omit the details.

Returning to the proof of the lemma, define  $\delta = \sum_{i=1}^{3n-2} c_i e_i$ . By (1) and the last claim it follows that  $\|\delta\|_2^2 = \sum_{i=1}^{3n-2} c_i^2 = O(n/d)$ . On the other hand,  $y - \delta$  is a linear combination of  $e_{3n-1}$  and  $e_{3n}$ .  $\square$

The vectors  $x - \epsilon$  and  $y - \delta$  are independent since they are nearly orthogonal. Indeed, if  $\alpha(x - \epsilon) + \beta(y - \delta) = 0$ , then  $\alpha x + \beta y = \alpha \epsilon + \beta \delta$ , and so  $6n\alpha^2 + 2n\beta^2 = \|\alpha \epsilon + \beta \delta\|_2^2$ . But

$$\begin{aligned} \|\alpha \epsilon + \beta \delta\|_2 &\leq |\alpha| \|\epsilon\|_2 + |\beta| \|\delta\|_2 \\ &= O((|\alpha| + |\beta|)\sqrt{n/d}). \end{aligned}$$

Thus  $6\alpha^2 + 2\beta^2 = O((\alpha^2 + \beta^2)/d)$ , and hence  $\alpha = \beta = 0$ .

Therefore, by the above lemma, the two vectors  $\sqrt{3ne_{3n-1}}$  and  $\sqrt{3ne_{3n}}$  can be written as linear combinations of  $x - \epsilon$  and  $y - \delta$ . Moreover, the coefficients in these linear combinations are all  $O(1)$  in absolute value. This is because  $x - \epsilon$  and  $y - \delta$  are nearly orthogonal, and the  $l_2$ -norm of each of the four vectors  $x - \epsilon, y - \delta, \sqrt{3ne_{3n-1}}$  and  $\sqrt{3ne_{3n}}$  is  $\Theta(\sqrt{n})$ . More precisely, if one of the vectors  $\sqrt{3ne_{3n-1}}, \sqrt{3ne_{3n}}$  is written as  $\alpha(x - \epsilon) + \beta(y - \delta)$  then, by the triangle inequality,  $\|\alpha x + \beta y\|_2 \leq \Theta(\sqrt{n}) + |\alpha| \|\epsilon\|_2 + |\beta| \|\delta\|_2$  which, by a calculation similar to the one above, implies that  $6\alpha^2 + 2\beta^2 \leq 1 + O((\alpha^2 + \beta^2)/d)$ , and thus  $\alpha$  and  $\beta$  are  $O(1)$ . It follows that the vector  $t$  defined in the algorithm is also a linear combination of the vectors  $x - \epsilon$  and  $y - \delta$  with coefficients whose absolute values are all  $O(1)$ . Since both  $x$  and  $y$  belong to the vector space  $F$  defined in the proof of Proposition 2.1, this implies that  $t = f + \eta$ , where  $f \in F$  and  $\|\eta\|_2^2 = O(n/d)$ . Let  $\alpha_i$  be the value of  $f$  on  $W_i$ ,

for  $1 \leq i \leq 3$ . Assume without loss of generality that  $\alpha_1 \geq \alpha_2 \geq \alpha_3$ . Since  $\|\eta\|_2^2 = O(n/d)$ , at most  $O(n/d)$  of the coordinates of  $\eta$  are greater than 0.01 in absolute value. This implies that  $|\alpha_2| \leq 1/4$ , because otherwise at least  $2n - O(n/d)$  coordinates of  $t$  would have the same sign, contradicting the fact that 0 is a median of  $t$ . As  $\alpha_1 + \alpha_2 + \alpha_3 = 0$  and  $\alpha_1^2 + \alpha_2^2 + \alpha_3^2 = 2 + O(d^{-1})$ , this implies that  $\alpha_1 > 3/4$  and  $\alpha_3 < -3/4$ . Therefore, the coloring defined by the sets  $V_j^0$  agrees with the original coloring of  $G$  on all but at most  $O(n/d) < 0.001n$  coordinates.

### 3.2 The iterative procedure

Denote by  $H$  the subset of  $V$  obtained as follows. First, set  $H$  to be the set of vertices having at most  $1.01d$  neighbors in  $G$  in each color class. Then, repeatedly, delete any vertex in  $H$  having less than  $0.99d$  neighbors in  $H$  in some color class (other than its own.)

**Proposition 3.2** *Almost surely, by the end of the second phase of the algorithm, all vertices in  $H$  are properly colored.*

To prove Proposition 3.2, we need the following lemma.

**Lemma 3.3** *Almost surely, there are no two subsets of vertices  $U$  and  $W$  of  $V$  such that  $|U| \leq 0.001n$ ,  $|W| = |U|/2$ , and every vertex  $v$  of  $W$  has at least  $d/4$  neighbors in  $U$ .*

**Proof.** Note that if there are such two (not necessarily disjoint) subsets  $U$  and  $W$ , then the number of edges joining vertices of  $U$  and  $W$  is at least  $d|W|/8$ . Therefore, by a standard calculation, the probability that there exist such two subsets is at most

$$\begin{aligned} &\sum_{i=1}^{0.0005n} \binom{3n}{i} \binom{3n}{2i} \left(\frac{2i^2}{di/8}\right) \left(\frac{d}{n}\right)^{di/8} \\ &\leq \sum_{i=1}^{0.0005n} \left(\frac{3en}{i}\right)^{3i} \left(\frac{16ei}{n}\right)^{di/8} \\ &\leq \sum_{i=1}^{0.0005n} \left(\frac{3en}{i}\right)^{di/40} \left(\frac{16ei}{n}\right)^{di/8} \\ &= \sum_{i=1}^{0.0005n} (48e^2)^{di/40} \left(\frac{16ei}{n}\right)^{di/10} \end{aligned}$$

$$\begin{aligned}
&\leq \sum_{i=1}^{0.0005n} (48e^2(16e/2000)^2)^{di/40} \left(\frac{16ei}{n}\right)^{di/20} \\
&\leq \sum_{i=1}^{0.0005n} \left(\frac{16ei}{n}\right)^{di/20} \\
&= O(1/n^{\Omega(d)}).
\end{aligned}$$

□

By repeatedly applying the property asserted by the above lemma with  $U$  being the set of vertices of  $H$  whose colors in the end of the iteration  $i-1$  are incorrect, we deduce that the number of incorrectly colored vertices decreases by a factor of two (at least) in each iteration, implying that all vertices of  $H$  will be correctly colored after  $\lceil \log_2 n \rceil$  iterations. This is because if a vertex in  $H$  is colored incorrectly at the end of iteration  $i$  it must have more than  $d/4$  neighbors in  $H$  colored incorrectly at the end of iteration  $i-1$ . To see this, observe that any vertex of  $H$  has at most  $2(1.01d - 0.99d) = 0.04d$  neighbors outside  $H$ , and hence if it has at most  $d/4$  wrongly colored neighbors in  $H$ , it must have at least  $0.99d - d/4 > d/2$  neighbors of each color other than its correct color and at most  $d/4 + 0.04d$  neighbors of its correct color. This completes the proof of Proposition 3.2. We note that by being more careful one can show that  $O(\log_d n)$  iterations suffice here, but since this only slightly decreases the running time we do not prove the stronger statement here. □

The proof of the next lemma is a simple application of the Chernoff bounds, which we omit.

**Lemma 3.4** *There exists a constant  $\gamma > 0$  such that almost surely the following holds.*

(i) *For any two distinct color classes  $V_1$  and  $V_2$ , and any subset  $X$  of  $V_1$  and any subset  $Y$  of  $V_2$ , if  $|X| = 2^{-\gamma d}n$  and  $|Y| \leq 3|X|$ , then  $|e(X, V_2 - Y) - d|X|| \leq 0.001d|X|$ .*

(ii) *If  $J$  is the set of vertices having more than  $1.01d$  neighbors in  $G$  in some color class, then  $|J| \leq 2^{-\gamma d}n$ .*

**Lemma 3.5** *Almost surely,  $H$  has at least  $(1 - 2^{-\Omega(d)})n$  vertices in every color class.*

**Proof (sketch).** It suffices to show that there are at most  $7 \cdot 2^{-\gamma d}n$  vertices outside  $H$ . Assume for contradiction that this is not true. Recall that  $H$  is obtained by first deleting all the vertices in  $J$ , and then by a deletion process in which vertices

with less than  $0.99d$  neighbors in the other color classes of  $H$  are deleted repeatedly. By the last lemma  $|J| \leq 2^{-\gamma d}n$  almost surely. Consider the first time during the deletion process where there exists a subset  $X$  of a color class  $V_i$  of cardinality  $2^{-\gamma d}n$ , and a  $j \in \{1, 2, 3\} - \{i\}$  such that every vertex of  $X$  has been deleted because it had less than  $0.99d$  neighbors in the remaining subset of  $V_j$ . Let  $Y$  be the set of vertices of  $V_j$  deleted so far. Then  $|Y| \leq |J| + 2|X| \leq 3|X|$ . We therefore get a contradiction by applying Lemma 3.4 to  $(X, Y)$ . □

### 3.3 The third phase

We need the following lemma, which is an immediate consequence of Lemma 3.3.

**Lemma 3.6** *Almost surely, there exists no subset  $U$  of  $V$  of size at most  $0.001n$  such that the graph induced on  $U$  has minimum degree at least  $d/2$ .*

**Lemma 3.7** *Almost surely, by the end of the uncoloring procedure in Phase 3 of the algorithm, all vertices of  $H$  remain colored, and all colored vertices are properly colored, i.e. any vertex colored  $i$  belongs to  $W_i$ . (We assume, of course, that the numbering of the colors is chosen appropriately).*

**Proof.** By Proposition 3.2 almost surely all vertices of  $H$  are properly colored by the end of Phase 2. Since every vertex of  $H$  has at least  $0.99d$  neighbors (in  $H$ ) in each color class other than its own, all vertices of  $H$  remain colored. Moreover, if a vertex is wrongly colored at the end of the uncoloring procedure (i.e. if it is colored  $i$  and does not belong to  $W_i$ ), then it has at least  $d/2$  wrongly colored neighbors. Assume for contradiction that there exists a wrongly colored vertex at the end of the uncoloring procedure. Then the subgraph induced on the set of wrongly colored vertices has minimum degree at least  $d/2$ , and hence it must have at least  $0.001n$  vertices by Lemma 3.6. But, since it does not intersect  $H$ , it has at most  $2^{-\Omega(d)}n$  vertices by Lemma 3.5, leading to a contradiction. □

In order to complete the proof of correctness of the algorithm, it remains to show that almost surely every connected component of the graph induced on the set of uncolored vertices is of size at most  $\log_3 n$ . In the rest of this section we sketch a proof of this fact. We note that it is easy to replace the term  $\log_3 n$  by  $O(\frac{\log_3 n}{d})$ , but for our purposes the above

estimate suffices. Note also that if  $p = o(\log n/n)$  some of these components are actually components of the original graph  $G$ , as for such value of  $p$  the graph  $G$  is almost surely disconnected (and has many isolated vertices).

**Proposition 3.8** *Almost surely the largest connected component of the graph induced on  $V - H$  has at most  $\log_3 n$  vertices.*

**Proof (outline).** Let  $T$  be a fixed tree on  $\log_3 n$  vertices of  $V$ . Our objective is to estimate the probability that  $G$  contains  $T$  as a subgraph that does not intersect  $H$ , and show that this probability is sufficiently small to ensure that almost surely the above will not occur for any  $T$ . To simplify the notation, we let  $T$  denote the set of edges of the tree. Let  $V(T)$  be the set of vertices of  $T$ , and let  $I$  be the subset of all vertices  $v \in V(T)$  whose degree in  $T$  is at most 4. Clearly  $|I| \geq |V(T)|/2$ . Let  $H'$  be the subset of  $V$  obtained by the following procedure, which resembles that of producing  $H$  (but depends on  $V(T) - I$ ). First, set  $H'$  to be the set of vertices having at most  $1.01d - 4$  neighbors in  $G$  in each color class. Then delete from  $H'$  all vertices of  $V(T) - I$ . Then, repeatedly, delete any vertex in  $H'$  having less than  $0.99d$  neighbors in  $H'$  in some color class (other than its own.)

**Lemma 3.9** *Let  $F$  be any subset of edges with endpoints in  $V$ . As before, denote by  $E$  the set of edges of  $G$ . Let  $H(F \cup T)$  be the value of  $H$  in case  $E = F \cup T$ , and let  $H'(F)$  be the value of  $H'$  in case  $E = F$ . Then  $H'(F) \subseteq H(F \cup T)$ .*

**Proof (sketch).** First, we show that the initial value of  $H'(F)$ , i.e., that obtained after deleting the vertices with more than  $1.01d - 4$  neighbors in a color class of  $G$  and after deleting the vertices in  $V(T) - I$ , is a subset of the initial value of  $H(F \cup T)$ . Indeed, let  $v$  be any vertex that does not belong to the initial value of  $H(F \cup T)$ , i.e.  $v$  has more than  $1.01d$  neighbors in some color class of  $(V, F \cup T)$ . We distinguish two cases:

1.  $v \in V(T) - I$ . In this case,  $v$  does not belong to the initial value of  $H'(F)$ .
2.  $v \notin V(T) - I$ . Then  $v$  is adjacent to at most 4 edges of  $T$ , and so it has more than  $1.01d - 4$  neighbors in some color class in  $(V, F)$ .

In both cases,  $v$  does not belong to the initial value of  $H'(F)$ . This implies the assertion of the lemma, since the initial value of  $H'(F)$  is a subgraph of the initial value of  $H(F \cup T)$  and hence any vertex which will be deleted in the deletion process for constructing  $H$  will be deleted in the corresponding deletion process for producing  $H'$  as well.  $\square$

**Lemma 3.10**

$$\Pr[T \text{ is a subgraph of } G \text{ and } V(T) \cap H = \emptyset]$$

$$\leq \Pr[T \text{ is a subgraph of } G] \Pr[I \cap H' = \emptyset].$$

**Proof.** It suffices to show that

$$\begin{aligned} \Pr[I \cap H = \emptyset | T \text{ is a subgraph of } G] \\ \leq \Pr[I \cap H' = \emptyset]. \end{aligned}$$

But, by the last lemma,

$$\begin{aligned} \Pr[I \cap H' = \emptyset] &= \sum_{F: I \cap H'(F) = \emptyset} \Pr[E(G) = F] \\ &\geq \sum_{F: I \cap H(F \cup T) = \emptyset} \Pr[E(G) = F] \\ &= \sum_{F' : F' \cap T = \emptyset, I \cap H(F' \cup T) = \emptyset} \Pr[E(G) - T = F'] \\ &= \Pr[I \cap H = \emptyset | T \text{ is a subgraph of } G], \end{aligned}$$

where here  $F$  ranges over the subsets of edges with endpoints in  $V$  and  $F'$  ranges over subsets that do not contain  $T$ .  $\square$

Returning to the (outlined) proof of Proposition 3.8 we note, first, that by modifying the arguments in the proof of Lemma 3.5 one can show that almost surely each of the graphs  $H'$  (corresponding to the various choices of  $V(T) - I$ ) misses at most  $2^{-\Omega(d)}n$  vertices in each color class. Since  $H'$  is independent of the choice of  $I$  it is not difficult to show that this implies that the probability  $\Pr[I \cap H' = \emptyset]$  is at most  $2^{-\Omega(d|I|)}$ . Since  $|I| \geq |V(T)|/2$  and since the probability that  $T$  is a subgraph of  $G$  is precisely  $(d/n)^{|V(T)|-1}$  we conclude, by the last lemma, that the probability that there exists some  $T$  of size  $\log_3 n$  which is a connected component of the induced subgraph of  $G$  on  $V - H$  is at most  $2^{-\Omega(d \log_3 n/2)} (d/n)^{\log_3 n-1}$  multiplied by the number of possible trees of this size, which is

$$\binom{3n}{\log_3 n} (\log_3 n)^{\log_3 n-2}.$$



Therefore, the required probability is bounded by

$$\begin{aligned} \binom{3n}{\log_3 n} (\log_3 n)^{\log_3 n - 2} 2^{-\Omega(d \log_3 n/2)} \left(\frac{d}{n}\right)^{\log_3 n - 1} \\ = O(1/n^{\Omega(d)}), \end{aligned}$$

completing the proof.  $\square$

## 4 Concluding remarks

1. There are many heuristic graph algorithms based on spectral techniques, but very few rigorous proofs of correctness for any of those in a reasonable model of random graphs. Our main result here provides such an example. Another example is the algorithm of Boppana [5], who designed an algorithm for graph bisection based on eigenvalues, and showed that it finds the best bisection almost surely in an appropriately defined model of random graphs with a relatively small bisection width. See also [6] for a recent application of spectral techniques for finding approximate edge separators.
2. By modifying some of the arguments of Section 3 we can show that if  $p$  is somewhat bigger ( $p \geq \log^3 n/n$  suffices) then almost surely the initial coloring  $V_i^0$  that is computed from the eigenvectors  $e_{3n-1}$  and  $e_{3n}$  in the first phase of our algorithm is completely correct. In this case the last two phases of the algorithm are not needed. It can also be shown that if  $p > 10 \log n/n$  the third phase of the algorithm is not needed, and the coloring obtained by the end of the second phase will almost surely be the correct one.
3. We can show that a variant of our algorithm finds, almost surely, a proper coloring in the model of random *regular* 3-colorable graphs in which one chooses randomly  $d$  perfect matchings between each pair of distinct color classes, when  $d$  is a sufficiently large absolute constant. Here, in fact, the proof is simpler, as the smallest two eigenvalues (and their corresponding eigenspace) are known precisely.
4. The results easily extend to the model in which each vertex first picks a color randomly, independently and uniformly, among the three possibilities, and next every pair of vertices of dis-

tinct colors becomes an edge with probability  $p$  ( $> c/n$ ).

5. If  $G = G_{3n,p,3}$  and  $p \leq c/n$  for some small positive constant  $c$ , it is not difficult to show that almost surely  $G$  does not have any subgraph with minimum degree at least 3, and hence it is easy to 3-color it by a greedy-type (linear time) algorithm. For values of  $p$  which are bigger than this  $c/n$  but satisfy  $p = o(\log n/n)$ , the graph  $G$  is almost surely disconnected, and has a unique component of  $\Omega(n)$  vertices, which is called the *giant component* in the study of random graphs (see, e.g., [1], [3]). All other components are almost surely sparse, i.e., contain no subgraph with minimum degree at least 3, and can thus be easily colored in total linear time. Our approach here suffices to find, almost surely, a proper 3-coloring of the giant component (and hence of the whole graph) for all  $p \geq c/n$ , where  $c$  is a sufficiently large absolute constant, and there are possible modifications of it that may even work for all values of  $p$ . At the moment, however, we are unable to obtain an algorithm that provably works for all values of  $p$  almost surely.
6. Our basic approach easily extends to  $k$ -colorable graphs, for every fixed  $k$ , where here the relevant eigenvectors are those corresponding to the  $k - 1$  smallest eigenvalues.
7. The existence of an approximation algorithm based on the spectral method for coloring *arbitrary* graphs is a question that deserves further investigation (which we do *not* address here.) We have just been informed that Karger, Motwani and Sudan [11] have obtained very recently some results in this direction.

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